The Intrinsic Dimensionality of the Indian Stock Market: A Geometric Approach to Factor-Based Market Analysis

1 Problem Statement

The Indian equity market, represented by broad indices like the NIFTY 500, is a high-dimensional system comprising hundreds of individual securities. For an analyst, this presents a significant challenge: tracking and modeling the co-movements of 500+ companies is computationally intensive and statistically unreliable. The sheer volume of data, a phenomenon known as the "curse of dimensionality," makes it difficult to distinguish meaningful market signals from noise.

The central problem this thesis addresses is the analytical complexity of this high-dimensional market space. The goal is to develop and validate a method to **lower the dimension of the market analysis**, moving from a space of hundreds of individual companies to a much smaller space of core, underlying factors. This reduced-dimension model will enable a more efficient and insightful way to analyze and explain the primary drivers of the Indian stock market.

2 Theoretical Motivation: The Geometry of Market Data

Our intuition for geometry, honed in two or three dimensions, fails in high-dimensional spaces. The theoretical properties of such spaces provide a powerful motivation for our approach.

If the stock market were a truly random system—where each company's daily return is an independent random variable—the geometry of the market's data would exhibit bizarre, counter-intuitive properties:

- **Distance Concentration:** The Law of Large Numbers states that the average of many independent random variables converges to its expected value. Since the squared Euclidean distance between two daily return vectors (y and z) is a sum of hundreds of squared differences, it can be viewed as a sum of d independent random variables.

$$|y - z|^2 = \sum_{i=1}^{d} (y_i - z_i)^2$$

  According to the Law of Large Numbers, the probability of the sample average deviating from the true mean is bounded:

$$P\left(\left|\frac{x_1 + \cdots + x_n}{n} - E(x)\right| \geq \varepsilon\right) \leq \frac{\mathrm{Var}(x)}{n\varepsilon^2}$$

  As the dimension d (equivalent to n in the formula) grows large, this inequality guarantees that the average of the terms $(y_i - z_i)^2$ will be very close to its expectation. This forces the total distance to "concentrate" around a single value, making it geometrically impossible to distinguish a market crash from a normal trading day.

- **Near-Orthogonality:** In a high-dimensional random space, any two randomly chosen vectors are, with overwhelming probability, nearly orthogonal (at a 90-degree angle) to each other. In a financial context, this would imply that the market's collective movement on one day is almost completely uncorrelated with its movement on any other day, which contradicts observed phenomena like trends and volatility clustering.

The fact that we empirically observe structure in financial markets (e.g., market-wide rallies, sector-specific trends) is strong evidence that stock returns are **not** independent random variables. This non-randomness implies that the market's data does not fill the entire 500-dimensional space. Instead, its dynamics are constrained to a much lower-dimensional subspace.

This thesis is built on the hypothesis that this underlying low-dimensional structure can be identified and extracted, providing a simplified yet powerful representation of the market.

3 Proposed Methodology: Dimensionality Reduction

To identify this low-dimensional subspace, this thesis will employ a powerful matrix factorization technique for dimensionality reduction. The goal is to find the "best-fit k-dimensional subspace" for the cloud of data points representing daily market returns.

3.1 Geometric Interpretation of the Method

Geometrically, the chosen method identifies the optimal k-dimensional subspace that maximizes the sum of the squared projections of the data points onto it. By the Pythagorean theorem, this is mathematically equivalent to minimizing the sum of squared distances from the points to the subspace. The axes of this subspace are the **principal components**—the orthogonal directions that capture the maximum variance in the data.

3.2 Research Steps

The methodology will proceed in four key stages:

1. **Data Collection and Preparation:** Historical daily adjusted closing prices for the constituents of the NIFTY 500 index will be collected over a significant period (e.g., 10 years). This price data will be transformed into a standardized log-return matrix. Standardization (Z-score normalization) is crucial to ensure that each stock contributes to the analysis based on its correlations, not just its volatility.

2. **Matrix Factorization:** A matrix factorization method will be performed on the standardized return matrix. This decomposition will break down the market's historical data into a set of orthogonal vectors that define the principal directions of variation and a set of corresponding values that quantify the importance of each direction.

3. **Determining the Market's Intrinsic Dimension:** The values obtained from the factorization, which represent the amount of variance captured by each principal component, will be analyzed using a **scree plot**. The plot will empirically demonstrate how much of the market's total movement can be explained by a small number of components. The point where the plot "flattens" indicates the market's effective or intrinsic dimensionality, k.

4. **Constructing the Low-Dimensional Model:** The market model will be reduced from d (approx. 500) dimensions to k dimensions by retaining only the top k principal components. These components represent the latent factors driving the market.

4 Expected Outcomes and Contribution

The primary outcome of this thesis will be a simplified, low-dimensional factor model of the Indian stock market. This directly addresses the problem of reducing the analytical space.

- **Market Explanation:** Instead of analyzing 500 individual stocks, the market's behavior can be explained through the lens of a few (k) principal components. The first component is expected to represent the overall market movement (the "market factor"), while subsequent components will represent other systematic risk factors, such as tensions between different industrial sectors or investment styles.

- **Analytical Efficiency:** This reduced model provides a computationally efficient framework for market analysis, risk assessment, and portfolio construction.

- **Novelty:** While dimensionality reduction is an established field, this thesis provides a novel framing by using the fundamental principles of high-dimensional geometry as the theoretical justification for the approach. It moves beyond a purely statistical application to a geometrically motivated one, creating a unique and rigorous foundation for the problem statement.

By bridging abstract geometric theory with a practical financial problem, this research will provide a clear and defensible method for simplifying the analysis of the complex Indian stock market.